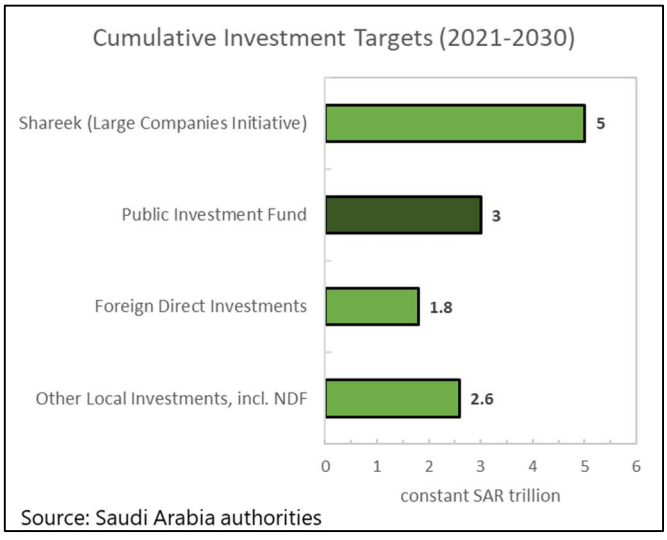
ASSESSING THE NATIONAL INVESTMENT STRATEGY USING THE DIGNAR-19 MODEL[1]

As part of its "Vision 2030", the Kingdom of Saudi Arabia unveiled in October 2021 its National Investment Strategy (NIS), which was designed to diversify the Saudi economy and raise potential GDP growth. Simulations conducted with IMF's DIGNAR-19 model assessed the potential impacts of NIS and find that the overall goals—when supported by fiscal reforms, labor supply reform, and higher public sector efficiency— boost potential non-oil growth by 4.8 percentage points to about 8.8 percent in the medium term.

1. Under "Vision 2030", the authorities have set out to ramp up both public and private investment, attract FDI and diversify the economy. The NIS is under the supervision of the Council of Economic and Development Affairs (CEDA) and is supported by four pillars, namely, 1) developing new investment opportunities, in strategic sectors 2) broadening the investment base, 3) removing financial barriers to private sector investment, and 4) improve the competitiveness of the business environment. These objectives are backed by more than 40 initiatives. After a first phase focused on overhauling the regulatory framework, efforts are now turning to tackling structural challenges and attracting investments.

2. Under the NIS, Gross Fixed Capital Formation (GFCF) is expected to grow by 13 percent per year until 2030. In nominal terms, this corresponds to a total of SAR 12.4 Tr cumulative new investments in the Saudi Economy by 2030 (or about twice what would be invested if investments were growing at the historical average). In support of this strategy, the Public Investment Fund (PIF), Saudi Arabia's Sovereign Wealth Fund, is expected to invest at least SAR 150 bn per year in the Saudi economy. Important contributions are also expected from the National Development Fund (NDF), whose consolidated assets reached about 17 percent of GDP in 2019. In addition, the NIS seeks to unlock private investment through the *Shareek* program, or Large Company Initiative, expected to generate SAR 5 Tr in investments. At the current juncture,



Source: Saudi Arabia authorities

[1] Authors: Flavien Moreau (FIN) and Zamid Aligishiev (RES). Authors thank Cian Ruane (RES) for inputs, useful suggestions, and comments.

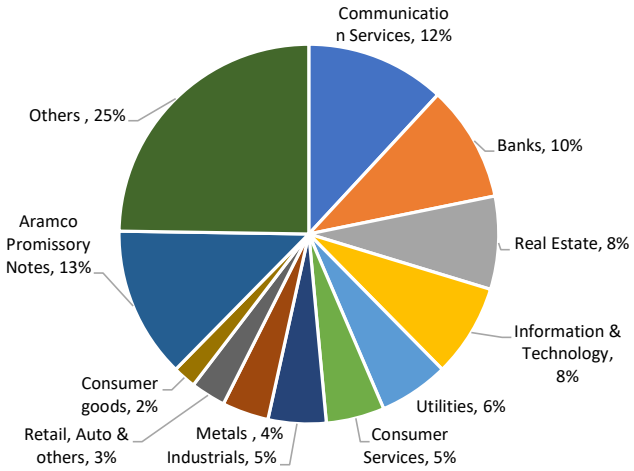
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diversifying the Saudi economy remains a challenge, as around 60 percent of current annual investments are concentrated in the hydrocarbon sector, as well as more than half of the FDI.[2]

Box 1. NIS and Strategic Sectors

The NIS seeks to stimulate private investments, including in key strategic sectors that the PIF has been tasked to help develop. These 13 strategic sectors include: Aerospace and Defense; Automotive; Transport and Logistics; Food and Agriculture; Construction; Entertainment, Leisure, and sports; Financial services; Real Estate, Utilities and Renewables; Metals and Mining; Health care; Consumer Goods and Retail; and Telecom, Media, and Technology. As of December 2020, some of these sectors were already well represented among PIF total assts holdings. While the catalytical effect on private sector is expected to vary, evidence suggests potential strong response from private companies in sectors such as Entertainment, Health, and Renewables (IMF, 2020).

PIF sectoral diversification



Source: Moody's

3. Using the DIGNAR-19 model calibrated to the Saudi economy, we find that simulations of the initiatives undertaken under the NIS can substantially raise non-oil potential growth. However, the magnitude of the increase and its sustainability are subject to several uncertainties, in particular, concerning the returns on capital efficiency of investment expenditure and the degree to which reforms will be successfully implemented and deliver the expected results. To quantify these uncertainties and highlight the role of the key assumptions, we formulate several scenarios and examine different calibrations of the model**.**

4. Simulations from the DIGNAR-19 model help quantify growth dividends from each channel as well as the impact on private investment DIGNAR-19 is a dynamic general equilibrium model (see Aligishiev, Melina and Zanna, 2021) produced at the IMF to deliver

[2] Inward FDI jumped from SAR 20.3 bn in 2020 to SAR 72.4 bn in 2021, of which SAR 46.5 bn correspond to the sale of a 49 percent equity stake in Aramco Oil Pipelines to a consortium led by EIG.

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quantitative macroeconomic assessments and policy scenario analyses in open developing economies.[3] The framework abstracts from the full input-output structure of the economy and incorporates three sectors of production (nontraded goods, non-resource traded goods and traded natural resources) as well as financially and non-financially constrained households.[4] The government has access to several fiscal instruments (productive and unproductive expenditures and various taxes) and both concessional and non-concessional debt, both domestic and external (see Annex I for key assumptions used for the calibration of the Saudi economy). We consider three scenarios:

- **Baseline**. This scenario is the one aligned with 38icroframework underpinning the July 2022 Article IV Consultations, which assumes that GFCF would keep growing at a rate of about 6 percent in nominal terms, with some partial implementation of the structural reforms under Vision 2030, delivering a medium-term non-oil GDP growth of 4 percent.

- **Moderate NIS investment scale up with stronger reform implementation**. This scenario is considering a richer set of policy changes, with fuller reform implementation, but a more measured pace of investment increases (9 percent on average on an annual basis after 2022, which, if sustained until 2030, would cover about 90 percent of the planned investments under the NIS). The reform scenario includes fiscal, labor, public investment, and regulatory reforms. Measures are assumed to be adopted in four areas: (1) a larger scale-up in combined public and PIF investment of 2.4 additional percent of GDP in 2022 and 5 percent of GDP in 2023-2026 ; (2) improvements in public investment efficiency is set to increase by 20 percentage points, following rationalization of procurements and other measure following PIMA guidelines (see Box 1); (3) fiscal reforms, including closing tax efficiency gap and rationalization of the high wage bill, including through strategic workforce planning and review, yield a gradual improvement of the fiscal balance of 0.5 percent of GDP by 2026; (4) labor market policies, boosting labor force participation, including of Saudi women, increases labor supply by 2 percentage points by 2026 relative to the baseline scenario.

- **NIS pure investment scale up**. This scenario assumes that the NIS investment targets are fully met and follow the 13 percent annual growth rate specified in the NIS. Implementation of structural reforms is assumed to be the same as in the baseline. Therefore, this scenario isolates the differential impact on non-oil potential output of the large NIS scale-up relative to the baseline.

[3] DIGNAR-19 is an extension of the Debt, Investment, Growth and Natural Resources (DIGNAR) model by Melina, Yang and Zanna (2014, 2016), which in turn is an extension of the Debt, Investment and Growth (DIG) model by Buffie and others (2012).

[4] See Baqaee Farhi (2019), Carvalho and Tahbaz-Salehi (2019) and Liu (2019) for recent advances in input-output macroeconomic modeling.

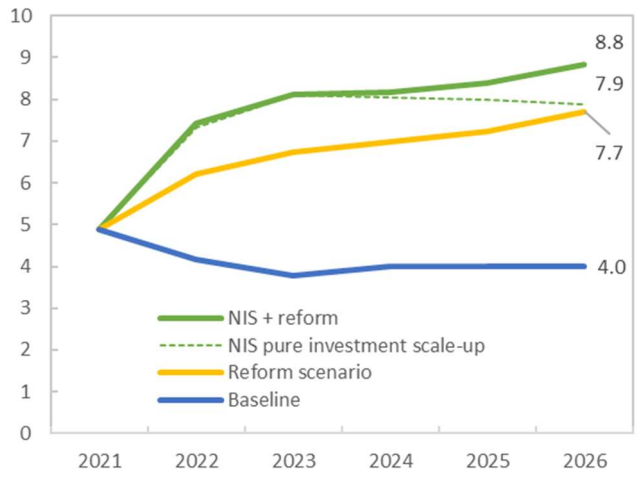
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- **NIS investment scale up combined with reforms**. This scenario also assumes that NIS investment targets are fully met and that, additionally, all the policies envisaged in the reform scenario are implemented.

5. The scenario with full NIS investment scale-up combined with reforms could raise non-oil potential output growth to 8.8 percent in 2026, compared to 4 percent in the baseline scenario used in the current macro framework. Importantly, the impact from the pure investment scale-up itself would be hump-shaped, peaking in 2023 and slowly declining after that. In contrast, the reform package would build over time and help achieve long-lasting growth. Even if the full investment targets were not fully met, simulations show that a package of reforms coupled with a moderate investment scale-up would still boost non-oil potential growth at a gradual and sustained pace, reaching 7.7 percent in 2026 (Figure 1).

Figure 1. Raising Non-oil GDP

(Percent, y-o-y change)

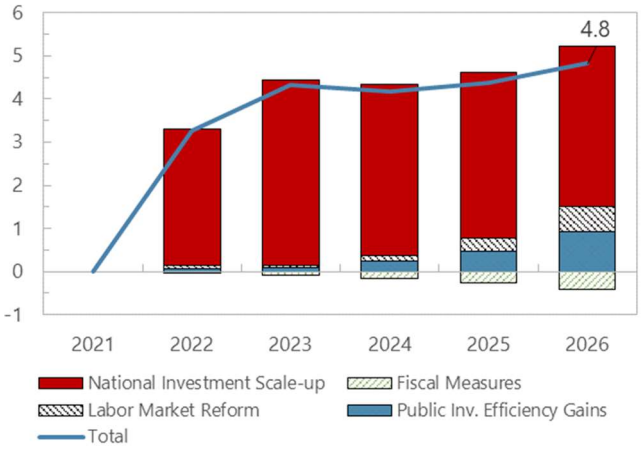


1/ The scenarios are computed using DIGNAR-19
Source: IMF staff estimates and projections.

6. Our results indicate that the combined effect of the reform package could deliver a medium-term increase in non-oil real GDP growth of 4.8 percentage points relative to baseline. The growth dividends build over time as reform indicators improve (Figure 2). While the steep ramp up in public spending, account for most of the potential output gain in the first years of the simulation horizon, backloaded dividends from labor market reforms and from gains in public investment efficiency boosts non-oil growth in the medium run, by about 0.5 and 0.8 percentage points respectively compared to the baseline. In 2026, dividends from the investment ramp up amount to 2.7 points, with a modest drag of 0.4 percentage point from fiscal measures.

Figure 2. Non-oil GDP Growth Dividends from Reforms

(Percentage points, relative to no-reform scenario)1/



1/ Reforms effects are simulated using DIGNAR-19.

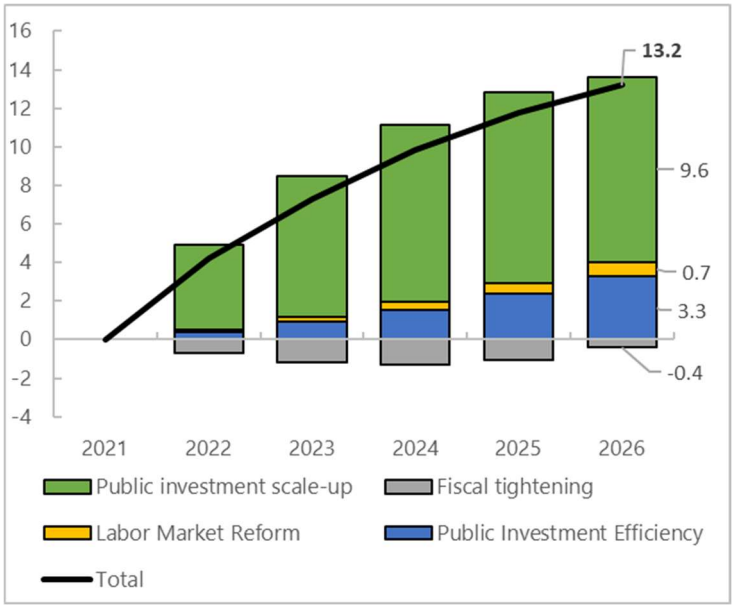
Source: IMF staff estimates and projections.

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7. The growth gains implied by the reform scenario would help mitigate the risks related to an overreliance on oil revenues while catalyzing private investments. Private investments are expected to increase by up to 13.2 percentage point relative to the baseline (Figure 3)**.** Simulations confirms the key role of the PIF's investments in driving additional private investments, while benefitting from positive spillovers from improvements on the labor market and in public investment efficiency. Under this scenario, and thanks to the modest fiscal consolidation, debt would stay at its current level and inflation and pressures on the real exchange rate would be contained, allowing the Saudi economy to rebalance towards a less oil-reliant economy and generate more opportunities for the private sector.

Figure 3. Impact on Private Investment

(Percentage points, relative to baseline)1/

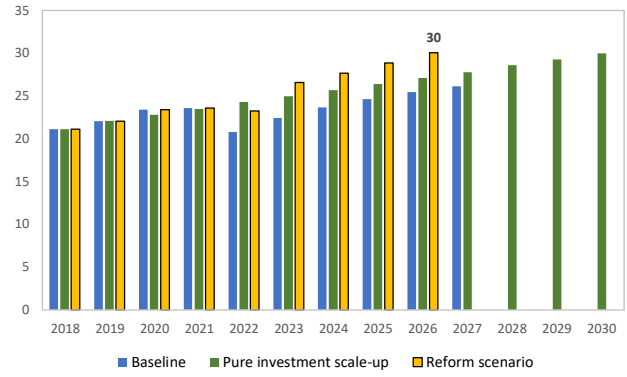


1/ Reforms effects are simulated using DIGNAR-19.
Source: IMF staff estimates and projections.

8. The Vision 2030 target of reaching an investment share of GDP of 30 percent would be reached earlier than 2030 in the reform scenario. While our main simulations anticipate a more modest annual increase in gross fixed capital formation (GFCF) of about 9 percent per year in nominal terms than expected under the NIS, GFCF as a share of GDP would rise faster than in the NIS forecasts, reaching 29 percent of GDP in 2026 and coming close to the range maintained by high-growth countries such as Korea or Singapore during the East Asian miracle. This faster pace is the result of discrepancies in investment multiplier assumptions, as, mechanically, lower multipliers result in higher GFCF shares for the same investment path.

Figure 4. Gross Fixed Capital Formation

(Share of GDP)



Sources: Saudi Arabia authorities, IMF staff estimates and projections.

9. From a quantitative standpoint, assumptions about investment multipliers therefore play an important role in driving these results. A wide range of multipliers can be found in the literature, reflecting a large heterogeneity in both countries' characteristics, the types of investment plans, and the economic context. Multipliers are higher when a country's current capital stock is low

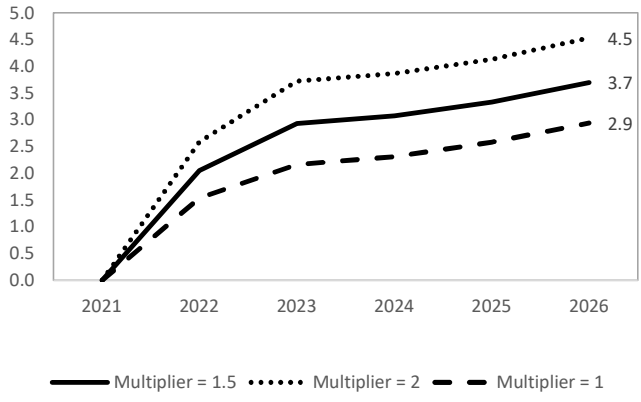
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(Izquierdo et al 2019) or when monetary policy is close to the zero lower bound. Absorptive capacity can also place a constraint on the efficiency or large investment scale-up (Adler 1965, Presbitero 2016). In addition, recent research suggests that multipliers can be higher for green investments for renewable energy projects [1.1 – 1.7] compared to [0.4 – 0.7] for fossil fuel projects (Batini et al 2021), confirming the importance of adequate project screening and benefits from diversification away from the oil sector.

10. To improve the robustness of the findings, we consider a range of multipliers. Investment multipliers are endogenous in the DIGNAR model, as they depend both on characteristics of the Saudi economy and estimated macroeconomic relationships. The main channels in the model include public investment efficiency (see Box 2), reallocation across sectors, adjustment costs, and the response of the private sector. In our preferred calibration, the model leads to a multiplier of 1.5, which sits at the middle of the range of public investment multipliers surveyed in Vagliasindi et al 2022. Given the uncertainty, we also consider alternative specifications leading to multipliers of 1 and 2, which provide a range for the impact of the reform package and how large a catalytic role can be played by the PIF or through *Shareek* programs.

11. Alternative specifications continue to support a sizeable effect on non-oil potential growth. Setting the public investment multiplier at 1 – that is, one riyal in investment delivers one additional riyal in output in the medium-run – delivers an additional boost of 2.9 percentage point in the reform scenario compared to the baseline (Figure 5). In contrast, if the multiplier is twice larger, the boost could reach 4.5 percentage points.

Figure 5. Impact on Private Investment
(Percentage points, relative to no-reform scenario)



Sources: IMF staff estimates and projections.

12. Another important source of uncertainty is the trajectory of oil prices, and the consequences on the Saudi economy, public finance, and the ability to stick to the investment plans. While no specific commitments have been made to direct additional fiscal resources from the central government to the PIF, authorities expect the PIF's assets under management to grow from around SAR 2.3 Tr currently, or about 60 percent of GDP, to SAR 4 Tr in 2025, or about 90 percent of GDP. Such a growth of assets under management, if generated organically, would require PIF to consistently deliver returns twice the size of the economy's growth rate. Alternatively, returns on investment in the range commonly observed would fall short of the announced target for PIF's assets under management. As a result, in order to properly assess the future path of the PIF's assets under management and available investment capacity in both upside and downside scenarios, more

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details on the PIF financial relationship with the central government and debt issuance plans would be required.[5]

Concluding Remarks

13. As projects are delivered, careful evaluation will allow to better assess the accuracy of the economic assumptions supporting the NIS. It will be important to assess the implicit multipliers associated with the various type of projects and account for uncertainty. Furthermore, these average multipliers are likely to exhibit substantial heterogeneity across projects, with Giga-projects carrying a high risk/ high return profile. Careful monitoring of projects through the PIF's Risk Committee and the NDF's new Impact Assessment Methodology will play an important role in maximizing medium-term risk-adjusted returns.

14. Policies to support innovation in key sectors could further lift potential output at longer horizons. While our simulations already find substantial gains, directing investments in R&D and technological upscaling in strategic sectors such as hydrogen or electric vehicles, may generate upside risks in the long run and complement improvements in the higher education system and human capital.[6] Towards such a goal, a national plan for Research, Development, and Innovation (RDI) was unveiled in July 2022, with a target of annual spending of 2.5 percent of GDP on research and development by 2040, with priorities including health, energy, and environmental sustainability.

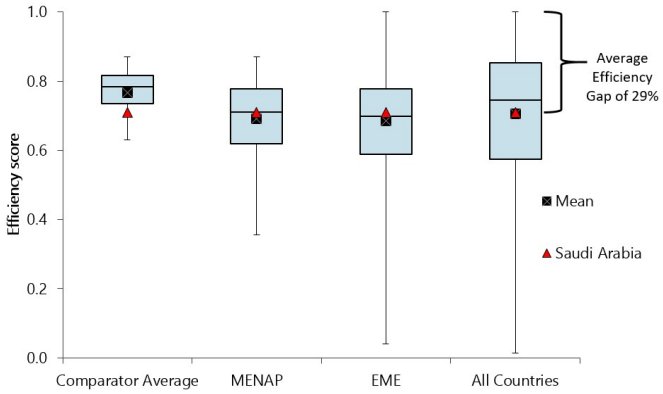
[5] For details on the implications of oil prices on public finances, see the Selected Issue Paper on the fiscal anchor. PIF has received its debut credit ratings in February 2022 and may consider tapping the market gradually by issuing green bonds.

[6] Myers and Lanahan (2022) find particularly large spillovers from publicly funded R&D in the energy sector.

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Box 2. Gains in Public Investment Efficiency

Public investment efficiency is the ability to translate a given level of spending into public goods, in particular economic and social infrastructure. On average, countries lose about one-third of their resources in the public investment process. As sources of inefficiencies are numerous and multidimensional Baum et al. (2020) builds a hybrid indicator accounting for both volume and quality of infrastructure and compute an efficiency frontier. With an efficiency of 0.71 measured over the past decade, Saudi Arabia faces significant gains from closing its efficiency gap. While there is no direct mapping between PIMA scores and the concept of public investment efficiency in the model (Gurara 2019), they can provide a good proxy for understanding the level of efficiency relative to peers and feasible room for efficiency gains.



Source. IMF, Fiscal Affairs Department and Baum, Tewodaj, and Verdier (2020)

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Annex I. The DIGNAR-19 Model

30. The Debt Investment Growth Natural Resources and COVID-19 (DIGNAR-19) model is a Dynamic General Equilibrium (DGE) model of a real small open economy developed at the International Monetary Fund to help country teams with quantitative macroeconomic assessments and policy scenario analysis in Low-Income Developing Countries (LIDCs) and Emerging Markets (EMs) during the COVID-19 pandemic. It is an extension of the DIGNAR model by Melina, Yang and Zanna (2014, 2016)[7] that accommodates the effects of the COVID-19 pandemic. Both DIGNAR and DIG models have gained a wide measure of acceptance for policy work. Over the past eight years, they have complemented the IMF and World Bank Debt Sustainability Framework (DSF) analysis, with over 65 country applications, in the context of Fund programs and surveillance work, and are now part of the IMF-WB Multipronged Approach for addressing debt vulnerabilities. These models helped inform policy analysis, based on qualitative and quantitative scenario analysis, on issues such as public investment surges, fiscal consolidations, PFM reforms and the collapse of commodity prices, among others. See Gurara, Melina and Zanna (2019) for an overview of DIG and DIGNAR applications.

31. The multiplier associated with the public investment program is determined by the magnitude of the investment surge, the source of financing, and the parameterization of the model. The model incorporates three main economic mechanisms for the analysis of public investment scale-ups: (i) a growth-investment nexus, (ii) a fiscal response required to maintain debt sustainability, (iii) and potential for both crowding-in and crowding-out of private investment. Public investment is assumed to be productive but has diminishing returns. Higher public capital raises productivity of private factors and increases real output through:

$$y_t = A_t(K_{t-1}^G)^{\alpha_g}(L_t)^{\alpha_N}(K_{t-1})^{1-\alpha_N}$$

where y_t is the real output of the economy; A_t is the total factor productivity; L_t is labor; K_t and K_t^G are private and public capital stock respectively. $\alpha_g \in (0,1)$ and $\alpha_N \in (0,1)$ govern the return on public capital and the labor share in production, respectively. The level of public capital at any given year is a sum of the stock of capital in the previous year, net of depreciation, and the new effective public investment expenditure. Formally:

$$\underbrace{K_t^G}_{\substack{New\ gov.\\ capital\ stock}} = \underbrace{(1-\delta)K_{t-1}^G}_{\substack{Old\ capital\ after\\ depreciation}} + \underbrace{\epsilon I_t^G}_{\substack{New\ investment\\ after\ waste}}$$

where I_t^G is the total public investment expenditure; $\delta \in (0,1)$ is the depreciation rate; and $\epsilon \in (0,1]$ governs the efficiency of public investment. Higher investment expenditure translates into higher capital stock, which in turn increases the marginal product of private capital. Higher marginal

[7] Which itself is an extension of the DIG model by Buffie and others (2012); Zanna and others (2019).

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product of capital incentivizes the private sector to match higher public expenditure with more private investments. The strength of this crowding in effect depends on the size of the private investment adjustment costs:

$$\underbrace{K_t}_{\substack{New\ capital \\ stock}} = \underbrace{(1-\delta)K_{t-1}}_{\substack{Old\ capital\ after \\ depreciation}} + \underbrace{I_t}_{\substack{New \\ investment}} \underbrace{-\frac{\kappa}{2}\left(\frac{I_t}{I_{t-1}}-1\right)^2 I_t}_{Adjustment\ cost}$$

where I_t is private investment expenditure and $\kappa \in [0,1)$ governs the size of the adjustment cost. Higher values of α_g and ϵ increase the size of the investment multiplier, while higher values of κ decrease it.

32. Calibrating DIGNAR-19 for Saudi Arabia. The DIGNAR-19 model is calibrated at an annual frequency using recent data capturing salient features of Saudi Arabia's economy. Table 1 reports the key parameter values necessary to pin down the initial steady state. Where possible, parameters are calibrated in line with data provided by the IMF country team. The initial rate of return on standard infrastructure investment is set at 25 percent, approximately in the middle of the range of estimates identified by the World Bank (2010).[8] The initial efficiency of public investment is set at 70 percent in line with most recent country assessment. Additional parameters are as in Aligishiev, Melina and Zanna (2020).

[8] "Cost-Benefit Analysis in World Bank Projects" (2010) by Independent Evaluation Group identifies a 17–31 percent range.

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Table 1. Calibrated Parameters and Initial Values for the Steady State
(In percent)

Definition	Value
Potential real GDP growth rate	3.2
Exports to GDP ratio	33.0
Imports to GDP ratio	16.9
Private investment to GDP ratio	13.3
Public investment to GDP ratio	12.1
Public consumption to GDP	20.0
Total public revenues to GDP ratio	40.0
Share of labor taxes in total public revenue	10.0
Share of consumption taxes in total public revenue	20.0
Share of government natural resource revenues in total government revenues	57.3
Public debt to GDP ratio	25.2
Private foreign debt to GDP ratio	53.8
Value added of the natural resource sector	40.0
Real interest rate on domestic debt	1.0
Initial return on public capital	25.0
Public Investment efficiency	70.0
Depreciation rate of public capital	7.0
Investment adjustment costs	5.0

Source: IMF staff calculations.

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